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Circulating exosome miRNA, is it the novel nutrient molecule through cross-kingdom regulation mediated by food chain transmission from microalgae to bivalve?

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A R T I C L E I N F O

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A B S T R A C T

MicroRNAs (miRNAs) can efficiently regulate gene expression at intracellular and extracellular levels. Plant-derived miRNAs are highly enriched in animal haemolymph and regulate mammalian gene expression. However, evidence for food-derived miRNAs in Mollusca species is lacking. In this study, we fed the microalga *Nannochloropsis oculata* to the pearl oyster *Pinctada fucata martensii* and detected dietary miRNAs in exosomes isolated from the haemolymph by RNA-seq. In total, 273 endogenous miRNAs were identified in all biological replicates. We identified 23 microalgae-derived miRNAs in the exosomes of pearl oyster haemolymph. Most microalgae-derived miRNAs showed high expression levels in both exosomes and microalgae and exhibited apparent variation among individuals. These food-derived miRNAs were predicted to participate in endocytosis, apoptosis, signal transduction, energy metabolism, and biomineralization by targeting multiple genes. These findings demonstrated the cross-kingdom transport of miRNAs from microalgae to bivalves and provide insights into novel nutrient transmission through the food chain.

1. Introduction

MicroRNAs (miRNAs) are members of a growing family of non-coding transcripts, are 21–23 nucleotides long, and regulate diverse biological processes, such as cell differentiation, neoplastic transformation, cell replication, and regeneration (Fabian et al., 2010). At the post-transcriptional level, they silence or activate specific protein-coding genes by targeting the 3′ or 5′ untranslated region of mRNA (Bartel, 2004; Bourin et al., 2011; Da Sacco and Masotti, 2013). miRNAs can efficiently regulate gene expression at intracellular and extracellular levels. For example, they can be transferred by extracellular vesicles (EVs), including microvesicles, exosomes, ectosomes, apoptotic bodies, and other extracellular vesicles, between cells. EVs are secreted by most cells and carry diverse cargoes, including proteins, RNA species, and lipids, between cells as a means of intercellular communication at both paracrine and systemic levels. Exosomes are 40–100 nm nano-sized vesicles with a cup-like shape; they are released from many cell types into the extracellular space and various body fluids (Mathivanan et al., 2010). By bearing the surface receptors/ligands of the original cells, exosomes could selectively interact with specific target cells and mediate intercellular communication by transporting DNA, RNA, proteins, lipids, and metabolites, thereby modulating recipient cells (Colombo et al., 2014; Rokad et al., 2019). The vesicle membrane of exosomes can protect miRNAs from degradation via extracellular RNase, resulting in greater stability than that of free miRNAs and promoting effective integration into specific recipient cells. Circulating miRNAs carried by exosomes that are derived from cells have been detected in breast milk, haemolymph, urine, and other samples. Changes in the levels of these circulating miRNAs are related to a variety of diseases, including type II diabetes, obesity, cardiovascular disease, cancer, and neurodegenerative diseases (Mitchell et al., 2008; Weber et al., 2010; Grasedieck et al., 2012). Exosomes also contribute to the pathogenesis of

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environmental chemical-induced chronic diseases. Misfolded proteins and miRNAs are examples of exosomal cargo molecules that can be altered by toxicants, ultimately changing the microenvironment of recipient cells and contributing to pathological processes (Rokad et al., 2019).

In 2012, Zhang et al. first discovered that miR168a in rice at high concentrations in the serum of Chinese subjects directly targets LDL receptor adaptor protein 1 (LDLRAP1) in hepatocytes, proving that miRNAs can regulate cells in distantly related taxa (Zhang et al., 2012a). Gene expression can be transmitted across species and play a regulatory role. The theory of miRNA "cross-kingdom regulation" predicts that miRNAs have powerful regulatory functions, with implications for co-evolution of broad taxa (e.g., microorganisms, plants, and animals), beyond horizontal gene transfer. Growing evidence shows that miRNAs can be transmitted from one species to another in a cross-kingdom manner. Honeysuckle-encoded atypical microRNA2911 directly targets influenza A viruses, even though a Chinese traditional decoction (Zhou et al., 2015). In insects, plant miRNAs are more enriched in beebread than in royal jelly and delay development and decrease body and ovary size in honeybees, thereby preventing larval differentiation into queens and inducing the development of worker bees (Zhu et al., 2017).

Mollusks are the second largest group of animals after arthropods. In marine ecosystems, bivalves form an important component of the food chain and are of considerable commercial value worldwide, with implications for food security. In bivalve species, EVs and their functions have been a focus of research. For example, exosome-like vesicles containing calcite crystals were observed at the mineralization front in the eastern oyster (Mount et al., 2004). In Pacific oyster, shell proteins that were likely deposited by exosomes were detected (Zhang et al., 2012b). Exosomes have been isolated from haemolymph or tissues, such as the mantle. They are involved in shell biomineralization and shell color formation (Mount et al., 2004; Chen et al., 2019; Chen et al., 2021). The roles of EVs in interactions between mollusks and microbes have been evaluated (Vanhove et al., 2015; Wang et al., 2018).

Bivalves have an open circulatory system, and their internal environment is highly susceptible to variation in the environment (Poelmann and Gittenberger-de Groot, 2019). As suspension feeders, they have highly developed processes for the cellular internalization of nano- and micro-scale particles (endo- and phagocytosis) integral to key physiological functions, such as intra-cellular digestion and cellular immunity (Canesi et al., 2012). Bivalve species filter water through the gills to feed on small plankton, such as algae and organic debris. Marine phytoplankton are responsible for 50% of the CO_2 that is fixed annually worldwide and contribute massively to other biogeochemical cycles in the oceans. However, there are many unresolved issues. For example, do these characteristics of bivalves make it easier to accept miRNAs from food sources for cross-species regulation? What roles do these cross-species miRNAs play in shellfish and algae? Does this cross-species regulation have a significant function in marine ecosystems?

Nannochloropsis has attracted sustained interest in algal biodiesel research owing to its high rate of biomass accumulation and high lipid content (Ma et al., 2014). *Nannochloropsis oculata* is an important microalga with robust physiological and biochemical characteristics; it can be easily cultured and is widely used in bivalve and rotifer feeding. Pearl oyster (*Pinctada fucata martensii*) is a sessile bivalve species distributed in northern China, India, Japan, and Southeast Asia. In this study, we detected miRNAs of pearl oysters feeding on *N. oculata* and identified microalgal miRNAs from the exosome-isolated miRNAs of pearl oyster. We evaluated whether diet-mediated miRNA transfer occurred between taxa. Our findings provide insights into the horizontal transfer of small RNAs across species and provide a new perspective on interactions within the ecosystem.

2. Materials and methods

2.1. *Nannochloropsis oculata* culture and pearl oyster feeding

N. oculata was purchased from SOUTH RANCH (China) and was cultured in sea water. Additionally, 1.5-year-old pearl oysters were cultured in 300 L breeding barrels for 2 weeks. Pearl oysters were fed cultured *N. oculata* thrice a day.

2.2. Isolation exosome from pearl oyster haemolymph

A blood sampling needle and EDTA anticoagulation tube were used to draw haemolymph from the adductor muscle without pollution. The sample was centrifuged at 1900 $\times g$ for 10 min with a bucket rotor at 4 °C to carefully collect the supernatant, i.e., the plasma, and the final 500 μL was discarded. The plasma was centrifuged again at 4 °C with 3000 $\times g$ for 15 min. Referring to the method described by Chen et al. (2021), plasma samples from five individuals were mixed for the isolation of exosomes as follows. Samples were centrifuged at 2000 $\times g$ and 4 °C for 30 min. The supernatant was carefully transferred to a new centrifuge tube and centrifuged again at 10,000 $\times g$ and 4 °C for 45 min to remove large vesicles. The supernatant was filtered through a 0.45 μm membrane and the filtrate was collected, transferred to new centrifuge tubes, and centrifuged with an ultra-speed rotor at 100,000 $\times g$ for 70 min at 4 °C. The supernatant was resuspended in 10 mL of pre-cooled 1× PBS, and an ultra-speed rotor was used for ultracentrifugation at 4 °C and 100,000 $\times g$ for 70 min. The supernatant was resuspended with 320 μL of pre-chilled 1× PBS, from which we obtained 20 μL for analyses by electron microscopy. Exosomes were stored at −80 °C.

2.3. Observation of exosomes

We obtained 10 μL of exosomes by pipetting 10 μL of the sample and dropping it on the copper net to settle for 1 min. The filter paper absorbed the floating liquid. Uranyl acetate (10 μL) was added dropwise to the copper mesh for precipitation for 1 min, and the floating liquid was absorbed by the filter paper. The sample was dried for a few minutes at room temperature. Electron microscopy (100 kV) and imaging were performed. A Nanoparticle Tracking Analysis (NTA) was performed. The expression of CD9 in exosomes was detected by western blotting with a CD9 antibody (Cat No: 60232-1-Ig).

2.4. RNA isolation and sequencing

Total RNA was isolated from feeding *N. oculata* by TRIzol. Then, the gel was cut following separation by agarose gel electrophoresis to select fragments of 18–30 nt. The exosomal miRNAs were isolated using the miRNA Isolation Kit (Omega Bio-Tek: R6727). The 3′ adaptor and 5′ adaptor were connected and RNAs of 36–48 nt were enriched. Then, reverse transcription and PCR amplification were performed on the small RNA with adaptors on both sides. Finally, the 140–160 bp products were recovered and purified by agarose gel electrophoresis to complete the library construction. Libraries were sequenced using Illumina HiSeq Xten by Gene Denovo Biotechnology Co. (Guangzhou, China).

2.5. Filtering of clean reads and alignment and annotation of small RNAs

Raw reads were filtered to obtain clean reads according to the following rules: low-quality reads containing more than one low quality (Q-value $\leq$ 20) bases or containing unknown nucleotides (N) were removed; reads without 3′ adapter or 5′ adapter were removed; reads containing adapters but no small RNA fragment or containing polyA sequences were removed; reads shorter than 18 nt were removed.

All clean reads were aligned with small RNAs in GenBank and Rfam databases to identify and remove rRNAs, scRNAs, snoRNAs, snRNAs, and tRNAs. In addition, all clean reads from the sequencing data for the three exosome samples were aligned against the reference genome of the pearl oyster *P. f. martensii* (CNGB Sequence Archive of China National GenBank DataBase; accession number CNP0002248). All clean reads for the three *N. oculata* samples were aligned against the reference transcriptome of *N. oculata.* Reads mapped to exons or introns or to repeat sequences were removed. All clean reads were then searched against the miRbase database for species identification based on known miRNAs. All of the unannotated reads were aligned against the reference genome of *P. f. martensii* or the reference transcriptome of *N. oculate* assembled in this study. According to the genome positions and hairpin structures predicted using miRDeep2, novel miRNA candidates were identified.

Expression levels of total miRNAs, including known and novel miRNAs, were calculated and normalized to transcripts per million (TPM). The formula was as follows: TPM = Actual miRNA counts / Total counts of clean reads $\times 10^6$.

2.6. Matching *N. oculata* miRNAs to exosomal miRNAs

Clean reads of exosomal small RNAs were blast searched to identify matches to miRNAs in *N. oculata* with 100% identity and identical lengths. The identified clean reads did not match to the genome sequence of the reference genome of *P. f. martensii.*

2.7. Target prediction

For pearl oyster samples, Miranda (Version 3.3a) and TargetScan (Version 7.0) were used to predict the targets of miRNAs. For *N. oculata* sample prediction, Patmatch (Version 1.2) was used to predict target genes.

2.8. Functional enrichment analysis of target genes

All target genes were mapped to KEGG and GO terms in the Kyoto Encyclopedia of Genes and Genomes database (https://www.kegg.jp/) and Gene Ontology database (http://www.geneontology.org/), respectively. Gene numbers were calculated for every term, and significantly enriched GO terms for DEGs in comparison with the genome background were defined by the hypergeometric test. The *P*-value was calculated as follows:

$$P = 1 - \sum_{i=0}^{m-1} \frac{\binom{M}{i}\binom{N-M}{n-i}}{\binom{N}{n}}$$

N is the total number of genes with a GO annotation; n is the number of DEGs in N; M is the total number of genes annotated to a certain GO term; and m is the number of DEGs in M. The calculated *P*-values were subjected to FDR correction, and FDR ≤ 0.05 was used as a threshold. A pathway enrichment analysis revealed significantly enriched metabolic pathways or signal transduction pathways for target genes compared with the whole genome background. The formula was the same as that used in the KEGG analysis.

2.9. Validation of microalgae-derived miRNAs in pearl oyster haemolymph exosome samples by qRT-PCR

Haemolymph exosomal miRNAs in *P. f. martensii* were isolated using the EasyPure® miRNA Kit (TransGen Biotech Co., Beijing, China). qRT-PCR was performed using TransScript® Green miRNA Two-Step qRT-PCR SuperMix Kit (TransGen Biotech Co.). Primers used for qRT-PCR are shown in Table S14. The relative expression levels were calculated by the $2^{-\Delta\Delta CT}$ method using U6 as a reference gene. Data are presented as means ± SD for three replicate reactions.

3. Results and analysis

3.1. Isolation and identification of exosomes from the haemolymph of pearl oysters

Exosomes from haemolymph were isolated by ultracentrifugation and observed by SEM. Haemolymph exosomes resembled typical disk-shaped vesicles (Fig. 1a). The average size of exosomes was 90.8 nm (Fig. 1b). The marker protein CD9 was detected by western blotting (Fig. 1c).

3.2. miRNA identification and target prediction from pearl oyster exosomes

Small RNA-sequencing for three exosome libraries generated approximately reads. The numbers of clean tags, annotated samples, and unannotated samples are shown in Table S1. According to the reference pearl oyster genome, miRNAs in exosomes were identified by Illumina sequencing. A total of 822 miRNAs were obtained, including 770 known miRNAs and 52 novel miRNAs (Table S2). The length distribution of small RNAs in the exosomes is shown in Fig. 2a. Fifty-nine known miRNAs could accurately predict the classical harpin structures with flanking sequences at their corresponding location in the genome (Table S3). Among these miRNAs, 273 were identified from all three biological replicates, with TPM ≥ 1 and count ≥ 3 (Table S4). More than 100 exosomal miRNAs were highly expressed in at least one sample with counts of more than five. Nearly 130 miRNAs had low concentrations, with counts of no more than two in all detected samples. This result indicated that there was variation in expression levels of individual miRNAs among samples. In these endo-exosomal miRNAs, 12 miRNAs were highly expressed with TPM values of higher than 10,000. According to the pearl oyster genome and gene annotation results (not published), a target prediction analysis revealed 1953 of miRNA: mRNA pairs. A functional enrichment analysis showed that "Endocytosis," "Autophagy," "mTOR signaling pathway," "Ubiquitin-mediated proteolysis," and "FoxO signaling pathway" were significantly enriched (Q $<$ 0.05) (Fig. 3, Table S5). Thirty-nine pathways were significantly enriched ($P < 0.05$). Most endo-miRNAs in haemolymph exosomes of *P. f. martensii* participated in the regulation of endocytosis and autophagy. The key genes in these pathways and related exosomal miRNAs were screened, as shown in Fig. 3 and Table S5. These genes participated in the regulation of cell proliferation and death, stress, energy metabolism, synthesis of unsaturated fatty acid, and protein synthesis and transport.

3.3. miRNA and mRNA identification from *N. oculata*

Small RNA-sequencing libraries for the feeding *N. oculata* were generated from three duplicate samples with approximately 9,333,057, 9,774,982, and 10,230,572 reads. The numbers of clean reads, annotated samples, and unannotated samples are shown in Table S6. The length distribution of the small RNAs is shown in Fig. 2. The most frequent lengths were 19 and 21 nt, followed by 22 nt. Finally, 182 known miRNAs and 131 novel miRNAs were identified in *N. oculata* (Table S7).

In this study, we sequenced the mRNAs from *N. oculata* for miRNA target prediction. A total of 64,990 unigenes were assembled with an N50 of 1296 bp (Fig. S1a). Transcriptome assembly completeness was evaluated by a BUSCO assessment, which indicated that 94.72% of the expected genes were completely present in the genome assembly (Fig. S1b); 69% of unigenes were annotated according to Nr, KEGG, KOG, and SwissProt databases.

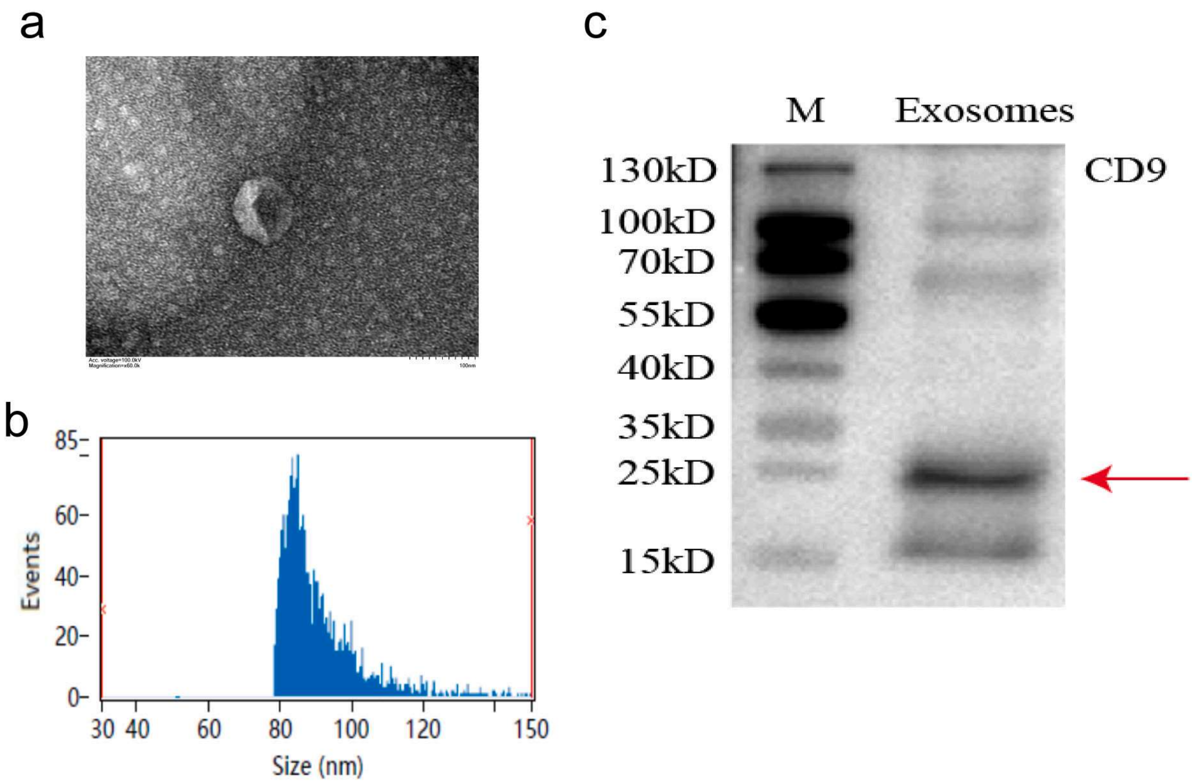

Fig. 1. Detection of isolated exosomes from haemolymph of *P. f. martensii*.
a. Haemolymph exosome detected by TEM; b. Particle size analysis of exosome samples by NTA; c. Western blot detection of CD9 of haemolymph exosomes.

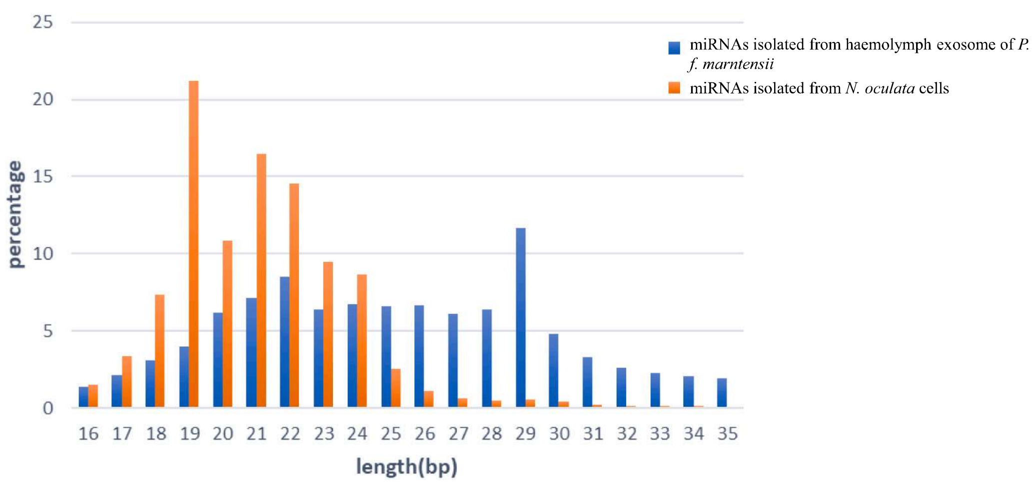

Fig. 2. Length distribution of endo-exosomal miRNAs and *N. oculata* miRNAs by RNA-seq.

3.4. Microalgae-derived miRNAs in pearl oyster haemolymph exosomes and target prediction in both pearl oyster and N. oculata

By using the clean reads produced from the exosomal small RNA library, reads were matched to 23 known miRNAs identified from *N. oculata*, as described above (Table 1). We validated four microalgae-derived miRNAs in haemolymph exosome samples of *P. f. martensii* by qRT-PCR (Fig. S2). In addition, the expression levels of nearly half of these microalgae-derived miRNAs were greater than median expression level of endo-exosomal miRNAs (Fig. 4). We analyzed their expression levels in *N. oculata* cells. Compared with the expression levels of microalgae-derived miRNAs in *N. oculata* cells, more than half of these miRNA expression levels exceeded the upper quartile of *N. oculata* miRNA expression (Fig. 5). Eleven miRNAs were among the top 50 highly expressed miRNAs in the detected *N. oculata* samples (Table 2).

To explore the roles of microalgae-derived miRNAs in pearl oyster, we predicted candidate target genes by a bioinformatic approach and then performed functional enrichment analyses. Nearly 700 target genes were found (Table S8). A Gene Ontology analysis showed that cytoskeleton structure, protein metabolism, phosphorylation modification, signal transduction, and glycogen and energy metabolism were significantly enriched (Table S9). A KEGG enrichment analysis revealed that the target genes were involved in endocytosis, FOXO signaling pathway, MAPK-mediated signal transduction, glycerophospholipid metabolism, and AMPK signaling pathway (Table S10 and 11, Fig. 6, Fig. S3). Typical genes that functioned in the regulation of biomineralization, such as

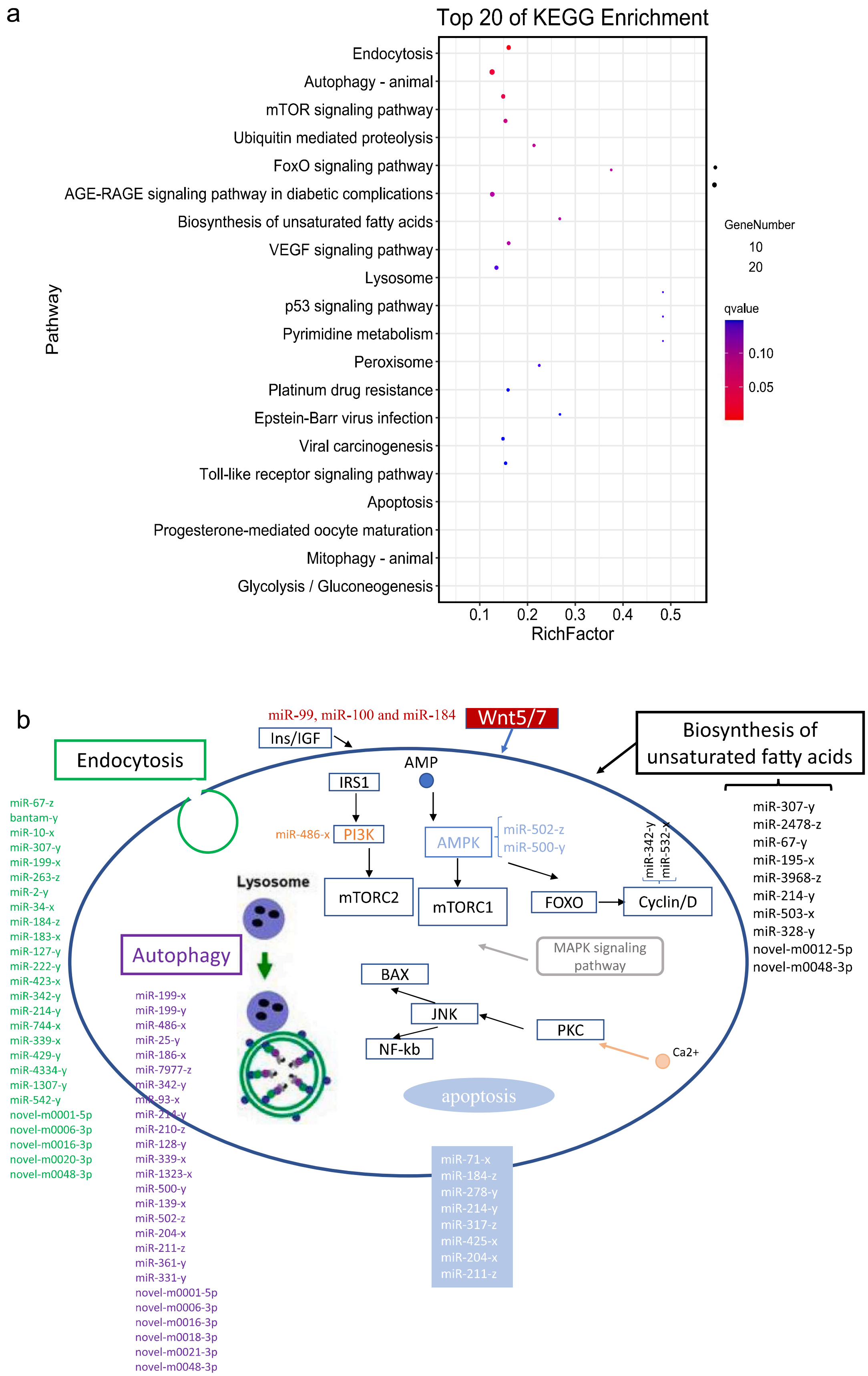

Fig. 3. KEGG enrichment analysis of target genes of exosomal miRNAs in *P. f. martensii*.
a. Top 20 significantly enriched pathways of exosome miRNA target genes in *P. f. martensii*. b. Exosomal miRNAs predicted to target genes involved in the enriched pathways, including endocytosis, autophagy, apoptosis, Wnt, mTOR, MAPK signaling pathways and biosynthesis of unsaturated fatty acids.

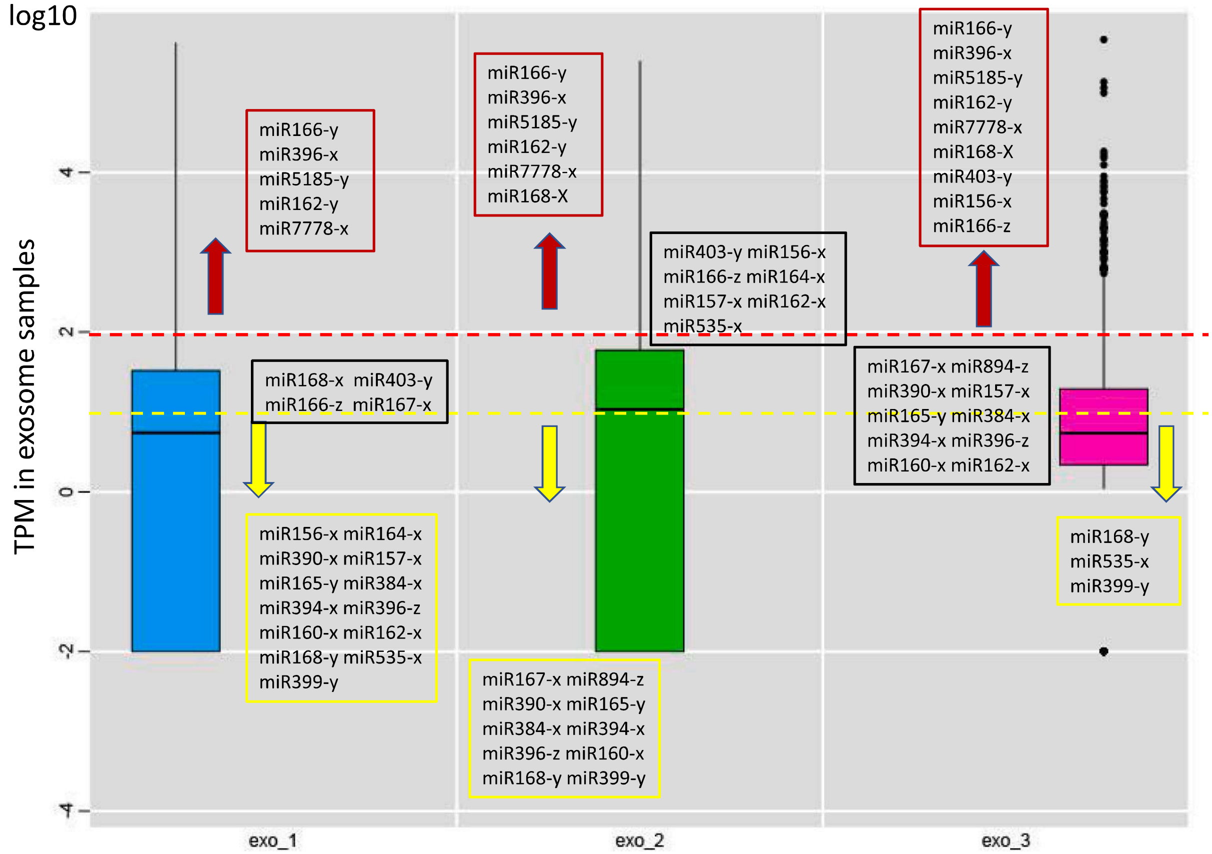

Fig. 4. Expression levels of "cross-species" microalgae-derived miRNAs and endo-exosome miRNAs (exo) in the haemolymph exosome samples by sRNA-seq in *P. f. martensii.*
Blue, green, and purple boxes represent the expression levels (TPM) of endo-miRNAs in three biological replicates of haemolymph exosome samples of *P. f. martensii*, exo_1, exo_2, and exo_3, respectively. The relative expression levels (TPM) of "cross-species" microalgae-derived miRNAs detected in haemolymph exosomes by sRNA-seq in red boxes exceed 2, those in the black box are between 1 and 2, and those in the yellow box are less than 1.

bone morphogenetic protein 2 (*BMP2*) and mothers against decapentaplegic homolog (*Smad*) in the TGF-beta signaling pathway, were targeted by microalgae-derived miRNAs (Fig. 6).

We predicted the potential targets of 23 microalgae-derived (*N. oculata*) miRNAs in their own cells. A total of 228 potential target genes were identified according to *N. oculata* transcriptome data (Table S12). A GO enrichment analysis showed that these miRNAs in *N. oculata* itself functioned in "ATPase activity," "pyrophosphatase activity," "hydrolase activity," "nucleoside-triphosphatase activity," and "histone acetyltransferase," which were closely related to energy metabolism and epigenetic regulation (Fig. 6, Table S13).

4. Discussion

4.1. Endo-miRNAs of exosomes have multiple functions in pearl oyster

miRNAs are involved in many biological processes in animals. They are small molecules with 18–31 nt and they widely exist in animals and plants. Compared with several thousands of mRNAs, only hundreds of miRNAs are detected in certain tissues or cells (Pritchard et al., 2012; Huang et al., 2019). Thus, RNA profiling by small RNA-seq is an effective approach. Exosomes serve as cargo transfer miRNAs in a circular system. These extracellular miRNAs act as a means of cell communication in the regulation of cell differentiation, development, and homeostasis. In this study, 822 miRNAs were detected from haemolymph exosomes; however, only 273 miRNAs were stably detected in all three replicates. Of the exosomal miRNAs, ~25% exhibited variation in expression. In humans, substantial differences in exosome-encapsulated microRNAs among individuals have been reported, and these are used as circulating biomarkers in disease detection and diagnosis (Joyce et al., 2016). The exosomal composition differs among cells of origin. For example, exosomes originating from prostate cancer cells contain certain miRNAs (miR-155b, 125b, and 130b) able to induce the neoplastic transformation of adipose-derived stem cells (Abd Elmageed et al., 2014). Environmental pollution also affects the biogenesis, secretion, and intercellular interactions of exosomes. For example, PCB quinone-stimulated p-MLKL enhances exosome biogenesis and secretion (Peng et al., 2021). Chronic PM2.5 exposure causes bronchial epithelial cells to undergo atypical hyperplasia and induces expression differences in exosomal miRNAs in vitro, and these changes are partially associated with tumorigenesis (Wang et al., 2021). Therefore, we proposed that such variation may reflect individual differences in health or physical conditions or a limited sequencing coverage.

All exosome-derived miRNAs need to enter host cells to exert their functions. Endocytosis is the most important mechanism for exosome uptake. For example, the intestinal transport of bovine milk exosomes is mediated by endocytosis in human colon carcinoma Caco-2 cells and rat small intestinal IEC-6 cells (Wolf et al., 2015). We found that the target genes of endo-exosomal miRNAs were apparently enriched in endocytosis and autophagy pathways, as determined by a KEGG pathway enrichment analysis. Endocytosis and autophagy are involved in transport by extracellular endocytosis and the degradation of intracellular organelles, respectively. Autophagy is a strictly regulated process involving the degradation and recycling of cell contents and contributes to organelle metabolism and reuse (Zhang et al., 2019). Exosomal miRNAs participate in the regulation of endocytosis and autophagy,

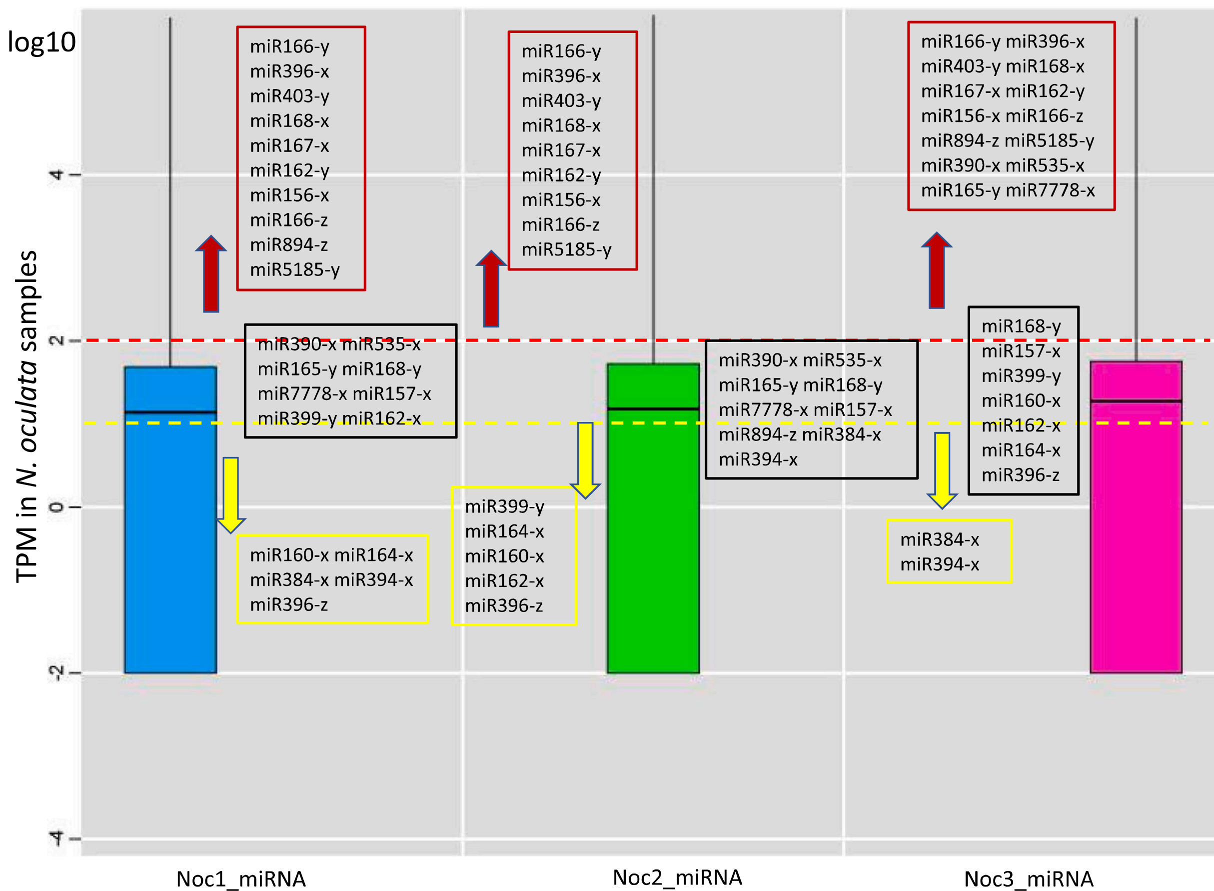

Fig. 5. Comparison of expression levels of microalgae-derived (*N. oculata*) miRNAs and *N. oculata* miRNAs (Noc) in the detected *N. oculata* samples. Noc1, Noc2, and Noc3 are the three biological replicates of *N. oculata* cell samples. Blue, green, and purple boxes represent the expression levels (TPM) of endo-miRNAs in Noc1, Noc 2, and Noc 3, respectively. The expression levels (TPM) of "cross-species" microalgae-derived miRNAs detected in haemolymph exosomes detected by sRNA-seq in *N. oculata* cells shown in the red box are greater than 2, those in the black box are between 1 and 2, and in the yellow box are less than 1.

thereby suggesting that they function in material uptake, energy metabolism, cell survival, and immune defense. mTOR promotes growth by stimulating the de novo synthesis of primary building blocks, such as proteins, nucleotides, and lipids, and by inhibiting autophagy (Shimobayashi and Hall, 2014). It could be activated by growth factors, such as insulin, amino acids, and wnt signaling, and in response to low energy (Liu and Sabatini, 2020). We found highly abundant endo-exosomal miRNAs, such as miR-99, miR-100, and miR-184, predicted to target *wnt5* and *wnt7*. In addition, target genes, such as *AMPK*, could affect the activation of the downstream FoxO signaling pathway improves cellular stress resistance (Salminen and Kaarniranta, 2012). The FoxO signaling pathway is also significantly enriched by endo-exosomal miRNA regulation, which is closely related to apoptosis, autophagy, and metabolism. Regulatory effects on the cell cycle and apoptosis during periods of environmental stress are critical for organismal survival. In this study, we found that endo-exosomal miRNAs could regulate apoptosis by targeting baculoviral IAP repeat-containing protein (BIRC) and ubiquitin ligases. Therefore, we proposed that these endo-derived circulating miRNAs have multiple functions in physiological regulation.

4.2. Variation in food-derived miRNAs abundant in haemolymph exosomes of pearl oyster

This is the first report of circulating miRNAs in the pearl oyster. The cross-kingdom transport between bivalves and dietary microalgae and regulatory effects were evaluated. Compared with endogenous microRNAs in humans, plant microRNAs are present at relatively lower levels but exhibit a high degree of variation among individuals (Zhang et al., 2019). We found 23 miRNAs derived from the microalga *N. oculata* by exosomal small RNA sequencing. We mixed exosome samples from three individuals, making it difficult to analyze differences among individuals. However, we detected heterogeneity in various miRNAs, such as miR168, miR403, miR156, and miR166. In principle, dietary microRNAs undergo processing, absorption by the gastrointestinal tract, delivery by the circulatory system, and transport to multiple tissues before exerting functions. However, plant miRNAs can further survive in the digestive system for >1 h without any significant decrease in concentrations (Philip et al., 2015). Compared with endo-exosomal miRNA expression levels, half of the *N. oculata*-derived miRNAs showed high expression. The microalgae-derived miRNAs with the highest expression (such as miR166y) were also the most abundant known miRNAs in *N. oculata* cells. This result indicated the high stability of microalgae-derived miRNAs in pearl oyster. We proposed that the intracellular digestion and open vascular system in Mollusca allowed substantial food-derived miRNAs to invade.

4.3. Functions of *N. oculata*-derived exosome miRNAs in donor *N. oculata* and bivalves

N. oculata is easily cultured and is a nutritious microalga widely used in bivalve culture. Recent studies have focused on microalgal miRNAs and their potential functions. Azaman et al. identified miRNAs from *Chlorella sorokiniana*, including 98 miRNAs in cultured microalgae subjected to normal and stress conditions (Azaman et al., 2020). Deng et al. (2017) found that miRNAs in *Botryococus braunii* are involved in metabolic and cellular processes, gene expression regulation, and stress/

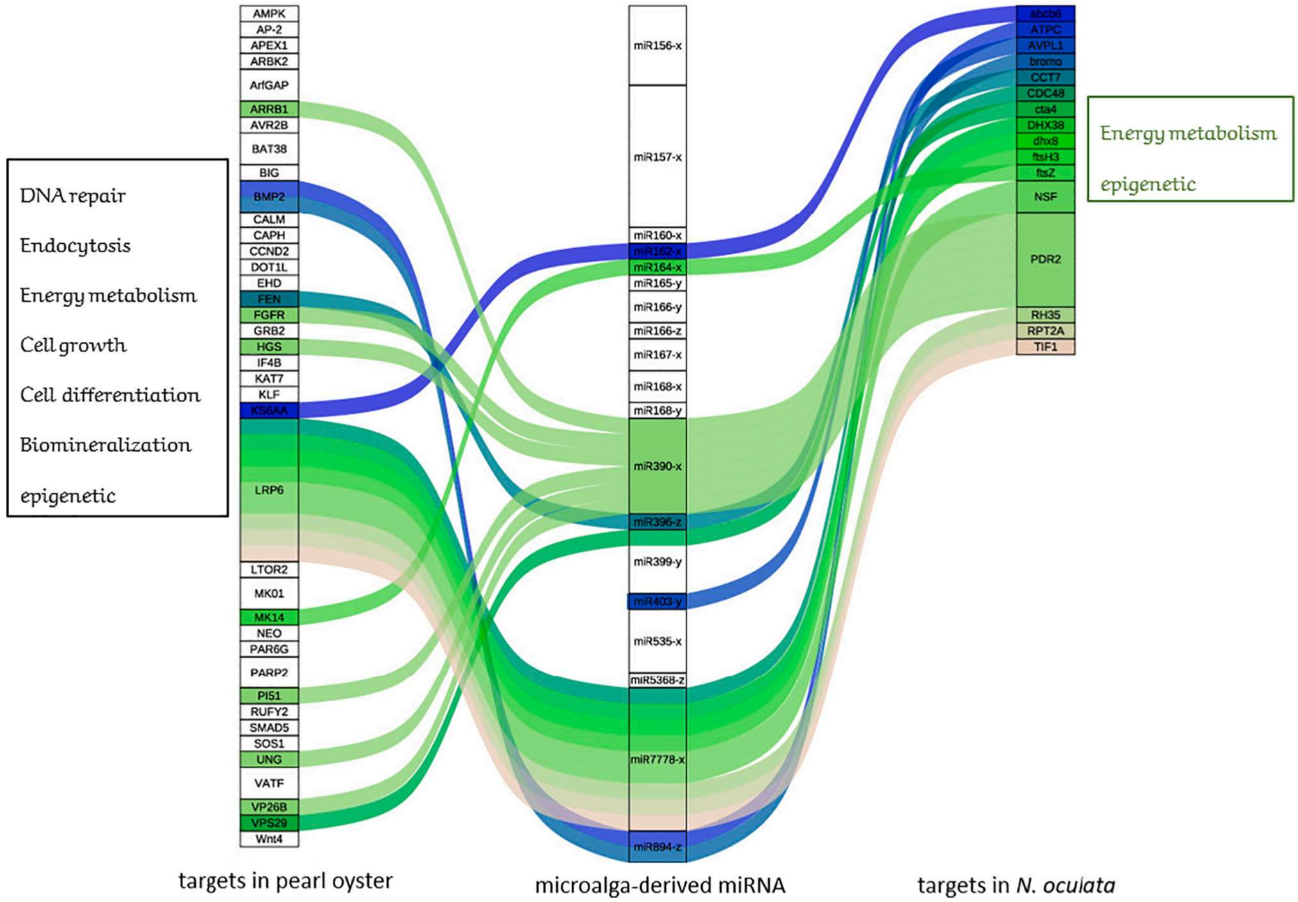

Fig. 6. Targets of microalgae-derived miRNAs in pearl oyster and microalgae *N. oculata* cells.
Microalgae-derived miRNAs from haemolymph exosomes of *P. f. martensii* (middle column), as well as candidate target genes by bioinformatic prediction in the host *P. f. martensii* (left column) and their derived microalgae *N. oculata* cells (right column) are shown.

Table 1
Sequenced, detected tag numbers and relative expression level of microalgae-derived (*N. oculata*) miRNAs in haemolymph exosome samples of *P. f. martensii.*

ID	Length	Seq	Total	exo_1_count	exo_2_count	exo_3_count	exo_1_TPM	exo_2_TPM	exo_3_TPM
miR166-y	21	TCGGACCAGGCTTCATTCCCC	35,364	1117	1506	32,741	3059.168	8157.65	35,565.31
miR396-x	21	TTCCACAGCTTTCTTGAACTT	6144	125	176	5843	342.3419	953.3508	6347.03
miR5185-y	18	GGAGGTTGGCTTAGAAGC	2363	526	740	1097	1440.575	4008.407	1191.63
miR162-y	21	TCGATAAACCTCTGCATCCAG	2013	50	75	1888	136.9368	406.2574	2050.863
miR7778-x	18	TCGAGTCGGCCTGCGCGG	1049	419	446	184	1147.53	2415.878	199.8723
miR168-x	21	TCGCTTGGTGCAGGTCGGGAA	525	9	21	495	24.6486	113.7521	537.6998
miR403-y	21	TTAGATTCACGCACAAACTCG	266	5	6	255	13.6937	32.5006	276.9969
miR156-x	20	TGACAGAAGAGAGTGAGCAC	178	2	5	171	5.4775	27.0838	185.7508
miR166-z	21	GCGGACCAGGCTTCATTCCCC	147	4	4	139	10.9549	21.6671	150.9905
miR164-x	21	TGGAGAAGCAGGGCACGTGCA	70	2	3	65	5.4775	16.2503	70.607
miR167-x	22	TGAAGCTGCCAGCATGATCTGA	60	6	1	53	16.4324	5.4168	57.5719
miR894-z	18	TTCACGTCGGGTTCACCA	54	1	0	53	2.7387	0.01	57.5719
miR390-x	21	AAGCTCAGGAGGGATAGCGCC	48	3	0	45	8.2162	0.01	48.8818
miR157-x	21	TTGACAGAAGATAGAGAGCAC	46	0	2	44	0.01	10.8335	47.7955
miR165-y	21	TCGGACCAGGCTTCATGCCCC	32	2	1	29	5.4775	5.4168	31.5016
miR384-x	20	TTGGCATTCTGTCCACCTCC	32	3	1	28	8.2162	5.4168	30.4153
miR394-x	20	TTGGCATTCTGTCCACCTCC	32	3	1	28	8.2162	5.4168	30.4153
miR396-z	21	TTCCACAGCTTTCTTGAACGT	23	2	1	20	5.4775	5.4168	21.7252
miR160-x	21	TGCCTGGCTCCCTGTATGCCA	15	1	0	14	2.7387	0.01	15.2077
miR162-x	21	GGAGGCAGCGGTTCATCGATC	13	0	2	11	0.01	10.8335	11.9489
miR168-y	21	CCCGCCTTGCATCAACTGAAT	9	0	0	9	0.01	0.01	9.7764
miR535-x	21	TGACAACGAGAGAGAGCACGC	8	0	6	2	0.01	32.5006	2.1725
miR399-y	21	TGCCAAAGGAGAGTTGCCCTG	2	0	0	2	0.01	0.01	2.1725

defense functions. In this study, we highlighted the potential functions of 23 *N. oculata*-derived exosome miRNAs in pearl oyster. The most highly expressed miRNA (miR166) was identified under normal conditions, and miR164, miR156, miR396, and miR399 were found in the stress-induced *C. sorokiniana* library (Azaman et al., 2020). In rice, miR164 regulates drought resistance by targeting NAC genes (Fang et al., 2014). By targeting the squamosa promoter binding-like gene, miR156 mediates the response of *Arabidopsis* to heat stress (Stief et al.,

Table 2
Sequenced, detected tag numbers and relative expression level of microalgae-derived (*N. oculata*) miRNAs in *N. oculata* cells.

ID	Length	Seq	Total	Q1_miRNA_count	Q2_miRNA_count	Q3_miRNA_count	Q1_miRNA_TPM	Q2_miRNA_TPM	Q3_miRNA_TPM
miR166-y	21	TCGGACCAGGCTTCATTCCCC	12,722	3783	1748	7191	26,198.61	13,226.69	45,299.92
miR396-x	21	TTCCACAGCTTTCTTGAACTT	7439	2164	804	4471	14,986.46	6083.673	28,165.2
miR403-y	21	TTAGATTCACGCACAAACTCG	970	235	94	641	1627.458	711.2752	4037.999
miR168-x	21	TCGCTTGGTGCAGGTCGGGAA	963	285	106	572	1973.725	802.0763	3603.331
miR167-x	22	TGAAGCTGCCAGCATGATCTGA	846	229	86	531	1585.906	650.7412	3345.051
miR162-y	21	TCGATAAACCTCTGCATCCAG	533	153	54	326	1059.579	408.6049	2053.647
miR5185-y	18	GGAGGTTGGCTTAGAAGC	151	14	84	53	96.9549	635.6076	333.8751
miR156-x	20	TGACAGAAGAGAGTGAGCAC	124	40	11	73	277.0141	83.2343	459.8657
miR166-z	21	GCGGACCAGGCTTCATTCCCC	119	28	14	77	193.9098	105.9346	485.0638
miR894-z	18	TTCACGTCGGGTTCACCA	52	17	9	26	117.731	68.1008	163.7878
miR390-x	21	AAGCTCAGGAGGGATAGCGCC	43	11	4	28	76.1789	30.267	176.3868
miR535-x	21	TGACAACGAGAGAGAGCACGC	38	7	6	25	48.4775	45.4005	157.4883
miR165-y	21	TCGGACCAGGCTTCATGCCCC	25	5	6	14	34.6268	45.4005	88.1934
miR168-y	21	CCCGCCTTGCATCAACTGAAT	24	9	3	12	62.3282	22.7003	75.5944
miR7778-x	18	TCGAGTCGGCCTGCGCGG	24	6	4	14	41.5521	30.267	88.1934
miR157-x	21	TTGACAGAAGATAGAGAGCAC	18	5	2	11	34.6268	15.1335	69.2948
miR399-y	21	TGCCAAAGGAGAGTTGCCCTG	9	2	1	6	13.8507	7.5668	37.7972
miR160-x	21	TGCCTGGCTCCCTGTATGCCA	6	0	0	6	0.01	0.01	37.7972
miR162-x	21	GGAGGCAGCGGTTCATCGATC	4	2	0	2	13.8507	0.01	12.5991
miR164-x	21	TGGAGAAGCAGGGCACGTGCA	4	0	0	4	0.01	0.01	25.1981
miR384-x	20	TTGGCATTCTGTCCACCTCC	4	0	4	0	0.01	30.267	0.01
miR394-x	20	TTGGCATTCTGTCCACCTCC	4	0	4	0	0.01	30.267	0.01
miR396-z	21	TTCCACAGCTTTCTTGAACGT	4	1	0	3	6.9254	0.01	18.8986

Note: Q1, Q2, Q3 represented three biological duplications of microalgae *N. oculate*.

2014). The development of genetically engineered plants with enhanced stress tolerance by modifying miRNA156 target nodes and promising transformation tools can improve crop adaptation to abiotic stress (Jerome Jeyakumar et al., 2020). miR396 mediated protection against photo-oxidative damage by brassinosteroids and pathogen-associated molecular pattern-triggered immune responses against fungal pathogens in *Arabidopsis* (Soto-Suárez et al., 2017; Wang et al., 2020). We found that microalgae-derived exosomal miRNAs in donor *N. oculata* could regulate genes related to ATPase activity, pyrophosphatase activity, hydrolase activity, nucleoside-triphosphatase activity, and histone acetyltransferase and were closely related to basic energy metabolism and epigenetic regulation. That is, the miRNAs related to the environmental stress response in *N. oculata* can be taken up and transported into the circulating system by food chain transfer.

In this study, we were interested in the functions of microalgae-derived exosomal miRNAs in pearl oyster. The functions of these food-derived exosomal miRNAs requires uptake by host cells in pearl oyster. Cells internalize EVs by fusion with the plasma membrane or by endocytosis, which can be categorized by the type of endocytotic process, i.e., according to the type of cell and its physiologic state, as well as ligands on the surface of the exosome that recognize receptors on the surface of the cell (Abels and Breakefield, 2016). Accordingly, the regulatory roles of microalgae-derived exosomal miRNAs in endocytosis suggested that they function in exosome uptake and endocytosis-dependent molecular absorption. KEGG and GO enrichment analyses both demonstrated the importance of exosomal miRNAs in phosphorylation/dephosphorylation-mediated signal transduction, such as the FOXO, MAPK, and AMPK signaling pathways. These signaling pathways always cross-linked and participate in cell proliferation and differentiation, affecting glycogen, fatty acid, and glycerophospholipid metabolism. For example, the FOXO signaling pathway is involved in the responses to a wide range of external stimuli, including growth factors, such as insulin/IGF (Brown and Webb, 2018). Phospho-AMPK phosphorylates several enzymes and transcriptional regulators, such as FoxO3, involved in metabolic regulation (including lipid metabolism and glucose uptake), to replenish the ATP supply (Hardie and Carling, 1997). FOXOs integrate signals from insulin/IGF signaling, AMPK, HDACs, and possibly other inputs to respond to changes in nutrient availability. In pearl oyster, insulin-like growth factors significantly increase primary mantle cell activity and induce the expression of proliferating cell nuclear antigen (PCNA) by activating the mitogen-activated protein kinase (MAPK) and phosphatidylinositol 3-kinase (PI3K) pathways (Zhang and He, 2020). mTOR, a serine/threonine kinase, is a master regulator of cellular metabolism and promotes cell growth in response to environmental cues. Insulin and IGF activate their cognate receptors and subsequently activate the PI3K/AKT signaling axis, the upstream regulator of mTORC1 (Kim and Guan, 2015). In this study, microalgae-derived exosomal miRNAs regulated the mTOR signaling pathway by targeting growth factor receptor-binding protein 2 (*GRB2*), low density lipoprotein receptor-related protein 5/6 (*LRP5,6*), *Wnt4*, and *MAPK*, which are core genes in the growth factor signaling axis, suggesting that "cross-species" miRNAs have regulatory roles in energy metabolism, cell growth, and cell differentiation. These findings are also consistent with the cross-kingdom regulatory effects of miRNAs derived from happy tree in focal adhesion, of lipolysis in adipocytes, and mTOR signaling pathways in cancer (Kumar et al., 2017).

In addition, *BMP2* and *Smad* in the TGF-beta signaling pathway were target genes of the newly identified miRNAs. In pearl oyster, *SMAD4* plays a role in biomineralization and can transduce BMP2 signals (Zhao et al., 2016). The microalgae-derived miRNAs could participate in the biomineralization in shell formation.

Despite a lack of enrichment for epigenetic pathways, several miRNAs transported between microalgae and bivalves, such as miR167-x, miR894-z, and miR535-x, target genes related to histone acetylation modification, thereby regulating chromatin modification. Epigenetic regulation could rapidly alter gene expression and responses to environmental change, thereby controlling phenotypic plasticity. We proposed that coevolution between microalgae and pearl oyster may contribute to environmental adaptation via exosomes carrying miRNAs.

4.4. Future directions

Extensive studies have investigated the impacts of environmental changes on marine bivalves and microalgae. However, very little is known about the interaction between microalgae and bivalves via the food web. Under global warming, ocean acidification and extreme weather cause changes in the nutrient composition of microalgae, which directly affects the growth of bivalve species. At the molecular level, gene exchange between species (i.e., cross-species transfer), including the exchange of miRNAs, could rapidly affect gene expression and physical conditions of hosts. miRNAs regulate the responses of microalgae to environmental changes. Our future research will focus on 1) the identification of environmentally sensitive miRNAs in microalgae and 2) the detection of cross-species transfer of environmentally sensitive miRNAs and their regulatory roles in bivalves.

In conclusion, this study is the first to identity the endo- and plant-derived miRNAs from haemolymph exosomes isolated from a bivalve species. A large number of endo-miRNAs were secreted into the circulatory system through exosomes. Endo-exosomal miRNAs participated in the regulation of cell proliferation and death, immunity and stress, energy metabolism, synthesis of unsaturated fatty acid, and protein synthesis and transport. The 23 *N. oculata*-derived miRNAs detected in the haemolymph exosomes of the pearl oyster *P. f. martensii* showed relatively high concentrations and apparent variation among individuals. Microalgae-derived miRNAs participated in cell growth and death, signal transduction, endocytosis, energy metabolism, and biomineralization by targeting multiple genes in *P. f. martensii*. They were also closely related to energy metabolism and environmental adaptation in *N. oculata* cells. Therefore, highly expressed miRNAs induced by environmental changes in microalgae can be taken up and transported into the circulatory system of bivalves by food chain transfer.

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CRediT authorship contribution statement

Zhe Zheng: Conceptualization, Funding acquisition, Writing – original draft. **Zhijie Xu:** Investigation, Visualization, Formal analysis. **Caixia Cai:** Investigation, Visualization, Formal analysis. **Yongshan Liao:** Resources. **Chuangye Yang:** Resources. **Xiaodong Du:** Project administration. **Ronglian Huang**: Investigation, Formal analysis. **Yuewen Deng:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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